

Aim

To create a forensically sound disk image using tools like FTK Imager or Guymager and verify its integrity using SHA-256 hashing.

Theory

Disk imaging is the process of creating a bit-by-bit copy of a storage device to preserve digital evidence without altering the original data. Forensic tools ensure that hidden, deleted, and slack space data are also captured. To maintain evidence authenticity, cryptographic hash functions such as SHA-256 are used. A hash value acts like a digital fingerprint; if even a single bit changes, the hash value differs. Matching hash values of the original disk and the image confirms data integrity and ensures admissibility in legal investigations.

Devices Misc Help				
Rescan				
Serial nr.	Linux device	Model	State	Size
	/dev/sda	VMware_Virtual_S	☐ Idle	86.0GB
	/home/kali/Experiment1_Integrity_check/hello.txt	Manually added special device	☐ Idle	82.0Byte

Acquire image of /home/kali/Experiment1_Integrity_check/hello.txt

File format

☐ Linux dd raw image (file extension .dd or .xxx) ☒ Split image files

☒ Expert Witness Format, sub-format Guymager (file extension .Exx) Split size 2047 MiB

Case number 1

Evidence number 1

Examiner Faizan

Description Experiment 1

Notes Checking integrity

Destination

Image directory ... /home/kali/Downloads/

Image filename (without extension) Experiment 1

Info filename (without extension) Experiment 1

Hash calculation / verification

☒ Calculate MD5 ☒ Calculate SHA-1 ☒ Calculate SHA-256

☐ Re-read source after acquisition for verification (takes twice as long)

☒ Verify image after acquisition (takes twice as long)

Cancel Duplicate image... Start